# Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Absolute Error and MAE

> Prepare Absolute Error and MAE: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Calculate MAE MSE and **RMSE**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on **training** targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

### Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

### Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

> Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

### Concept to explain

**Absolute error** measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

| Term | Meaning |
|---|---|
| absolute error | absolute value of actual minus predicted |
| MAE | mean of absolute errors |
| unit | measurement scale retained by MAE |
| cancellation | positive and negative errors offset |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

## Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on absolute and squared **loss**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

### Readiness answer

The **training** mean among all constant **prediction**s.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

### End goal for the pictorial case

By the end, I can explain **absolute error** and mae using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Absolute Error and MAE using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

### Question to explain

QUESTION Calculate residuals, **absolute error**s, and mean **absolute error**.

MODEL IDEA **Absolute error** measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17. Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

### Step 1

STEP 1 Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17.

### Step 2

STEP 2 Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

### Step 3

STEP 3 MAE = (2 + 1 + 3) ÷ 3 = 2 cm.

### Step 4

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

**Concrete teaching case**

### The teaching case

Actual=[3,5,9], predicted=[2,7,6] cm.

### The teaching case

Calculate MAE.

**Exact explanation to model**

### Why the result follows

**Absolute error**s=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

**Explain the added worked example**

## Explain the added worked example · 1

A **loss** summarizes error magnitude without positive and negative **cancellation**. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly.

**RMSE** takes the square root of MSE so it returns to the target **unit**. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, **unit**s, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines absolute and squared **loss**.

Residuals [-1, 2,-3] cm have **absolute error**s [1, 2, 3], so MAE=6/3=2 cm. Squared errors [1, 4, 9] sum to 14, so MSE=14/3 square cm and **RMSE**=sqrt(14/3), about 2.16 cm. Keep the square and the average in the correct order.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

### Key terms

| Term | Meaning |
| --- | --- |
| MAE | Mean absolute error |
| MSE | Mean squared error |
| RMSE | Square root of mean squared error |
| Loss | A numerical penalty for prediction errors |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

### Absolute and squared loss at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
| --- | --- | --- |
| Residuals | minus 1 plus 2 minus 3 | Signed differences |
| Absolute values | 1 2 3 | MAE 2 cm |
| Squares | 1 4 9 | MSE 14 divided by 3 |

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

**Give MSE and RMSE with units. Explain the steps or evidence behind your response.**

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Absolute Error and MAE

Compare absolute and squared error without losing track of **unit**s.

### The facts we will use

> **Supplied facts**
>
> Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/21

# Absolute Error and MAE

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Absolute errors

Absolute values1, 2, 3 give MAE6/3=2 cm.



## 2. Squared errors

Squares1, 4, 9 give MSE14/3 cm2, approximately = 4.67 cm2.



## 3. Root squared error

RMSE=sqrt(14/3), approximately2.16 cm.

**CHANGE ONE THING**

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Absolute errors | Absolute values1, 2, 3 give MAE6/3=2 cm. |
| 2. Squared errors | Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². |
| 3. Root squared error | RMSE=sqrt(14/3), approximately2.16 cm. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

**Change one thing in the pictured case**

### Change one thing

Change: Convert every residual to millimetres.

### Predicted result

**Prediction**: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

### Why the result follows

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

**Ask while showing the picture**

### Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Absolute Error and MAE, reduce support deliberately and check individual reasoning before discussing answers.

### Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

## Board sequence

### 1. Put the facts on the board

Write the given facts: Actual=[3,5,9], predicted=[2,7,6] cm.

### 2. State the decision to explain

Write the question to explain: Calculate MAE.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: **Absolute error**s=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

### Hold back the independent task

Keep for an independent attempt: Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.

## Classroom dialogue

### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided response: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm. Exit response: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Use the prerequisite question: Which constant minimizes training squared error? Check the explanation before proceeding. |
| 7 to 18 minutes | Read the opening and model the worked evidence. Ask students to identify mae in context. |
| 18 to 32 minutes | Complete W 1 and W 2 with support: compute mae; compute squared loss. Require the evidence, not only the final response. |
| 32 to 45 minutes | Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation. |
| Session 2 start to 8 minutes | Retrieve the distinction between mae and mse. Invite a counterexample to an overconfident claim. |

| Stage | Teacher action |
| --- | --- |
| 8 to 20 minutes | Discuss the application and complete W 3: Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE? |
| 20 to 35 minutes | Collect W4 independently before giving hints. Assess these outcomes: Calculate MAE MSE and RMSE; Explain how squaring changes sensitivity to large errors. Record any support used. |
| 35 to 45 minutes | Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding. |

### Reduce help deliberately

#### Completion practice

Completion task: Give MSE and **RMSE** with **unit**s.

#### Reasoning to listen for

Teacher check: Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

#### Pictorial practice question

Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

#### Reasoned teacher answer

Its **unit** is cm², not cm. **RMSE** is the corresponding square-root statistic with cm **unit**s. Changed-case check: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Changed-case reasoning: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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TEACHER GUIDE · 4 OF 9

## Run the weekly evidence lab

> Run the Absolute Error and MAE investigation and use the first unsupported step to choose a

> teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

> 1. Write the hypothesis and expected **evidence** before running.

> 2. MAE Lab: calculate each row, verify **unit**s and denominator, then compare model and baseline on identical examples.

> 3. Run one normal, one boundary, one missing, and one shifted case.

> 4. Calculate the metric from raw counts or row-level errors.

> 5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |

| Record | What to capture |
|---|---|
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average **absolute error**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

A stronger claim makes a stronger project: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |

| What the response shows | Next teaching action |
| --- | --- |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Absolute Error and MAE questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

**Absolute error**s=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

### Independent question

#### Independent question · withhold the key

Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute**

**error**s and MAE.

### Expected independent answer

#### Teacher key · Expected independent answer

**Absolute error**s are 1, 2 and 3 cm. MAE is (1+2+3)/3 = 2 cm. Signed errors must not cancel.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Averaging signed errors is the same as mean **absolute error**.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Residuals=[-2,3,-1], summing to 0. **Absolute error**s=[2,3,1], MAE=6/3=2 in the target's

**unit**s. Signed errors can cancel even when every **prediction** is wrong.

### Explanations for the additional workbook examples

#### W1 Compute MAE

**Workbook question · W1**

Calculate MAE for the worked residuals.

**Teacher reasoning · W1**

Teacher explanation: (1+2+3)/3=2 cm.

**Teacher reasoning · W1**

MAE uses absolute residual sizes, so add 1, 2 and 3 and divide by three: (1+2+3)/3 = 2 cm. Positive and negative signed residuals must not cancel in this calculation.

State centimetres because **absolute error** retains the target **unit**.

#### W2 Compute squared loss

**Workbook question · W2**

Give MSE and **RMSE** with **unit**s.

**Teacher reasoning · W2**

Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

### Teacher reasoning · W2

Square the residual sizes to get 1, 4 and 9; their mean is 14/3 square centimetres. Take the square root to obtain **RMSE** sqrt(14/3), approximately 2.16 cm. MSE and **RMSE** describe related quantities but have different **unit**s.

## W3 Compare patterns

### Workbook question · W3

Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

### Teacher reasoning · W3

Teacher explanation: Both absolute sums are 6. Squared sums are 36 and 12, so MSE 12 versus 4.

## W4 Apply independently

### Workbook question · W4

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

### Teacher reasoning · W4

Teacher explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

### W5 Choose honestly

#### Workbook question · W5

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

#### Teacher reasoning · W5

Teacher explanation: Keep the declared primary metric and disclose any additional metric and reason. Report the large error and complete evaluation.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20.

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

#### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Absolute Error and MAE to Responsible Plant Classifier, adapt support and plan a later changed-case check.

#### Responsible Plant Classifier

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

#### Support and stretch

Use a tiny labelled dataset, explicit **unit**s and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 22: Train a Regression Model. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use mae and mse in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: MAE 2 m; MSE 20/3 square m; **RMSE** about 2.58 m. In cm, MAE 200, MSE 200000/3, **RMSE** about 258.20. Fresh retry residuals [-1, 1, 4]: MAE 2, MSE 6, **RMSE** sqrt 6 about 2.45 in corresponding **unit**s.

Require **unit**s and operation order. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: **Fitting** a line. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

### Plan the delayed return

Suggested return points in this level: ai-4/22, ai-4/24, ai-4/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |

| Record | Teacher evidence |
| --- | --- |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Absolute Error and MAE lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Calculate MAE MSE and **RMSE**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Give MSE and **RMSE** with **unit**s.

### Supported practice answer

Reasoned practice answer: Teacher explanation: MSE 14/3 square cm; **RMSE** sqrt(14/3) about 2.16 cm.

Check every learner before discussion: Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the relevant counts, **unit**s, procedure or saved record. For this week: (1+2+3)/3=2 cm.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Convert every residual to millimetres. Require a **prediction** and a reason before comparing. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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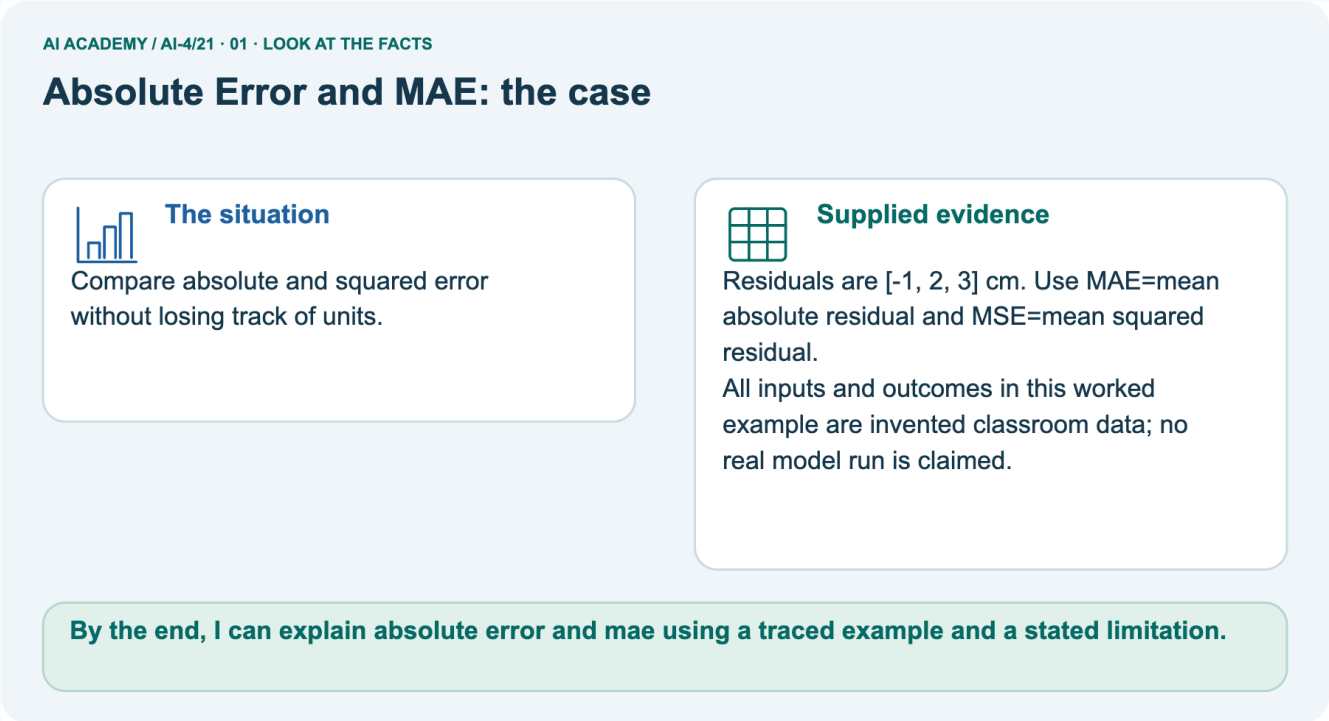
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## Visual concept journey

> Show one Absolute Error and MAE graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Absolute Error and MAE: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

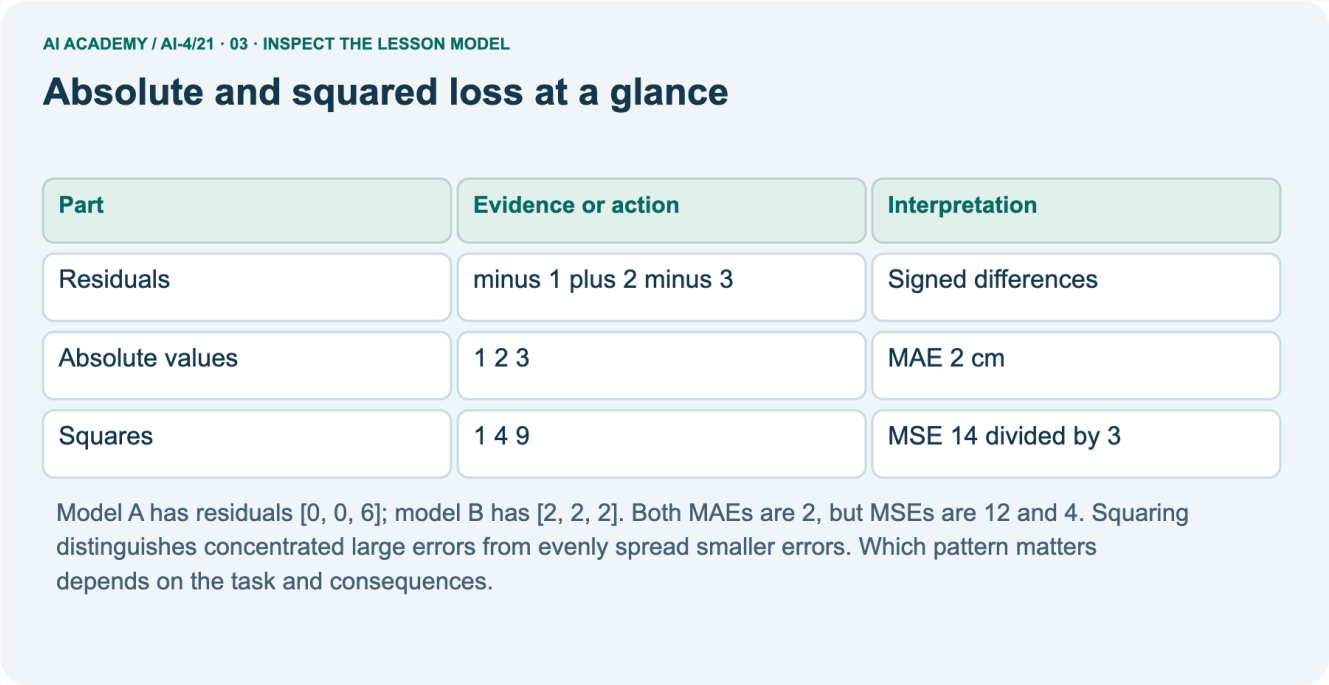
**Absolute error**s: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Absolute and squared loss at a glance




Follow the labelled picture with the matching explanation. Open the vector diagram

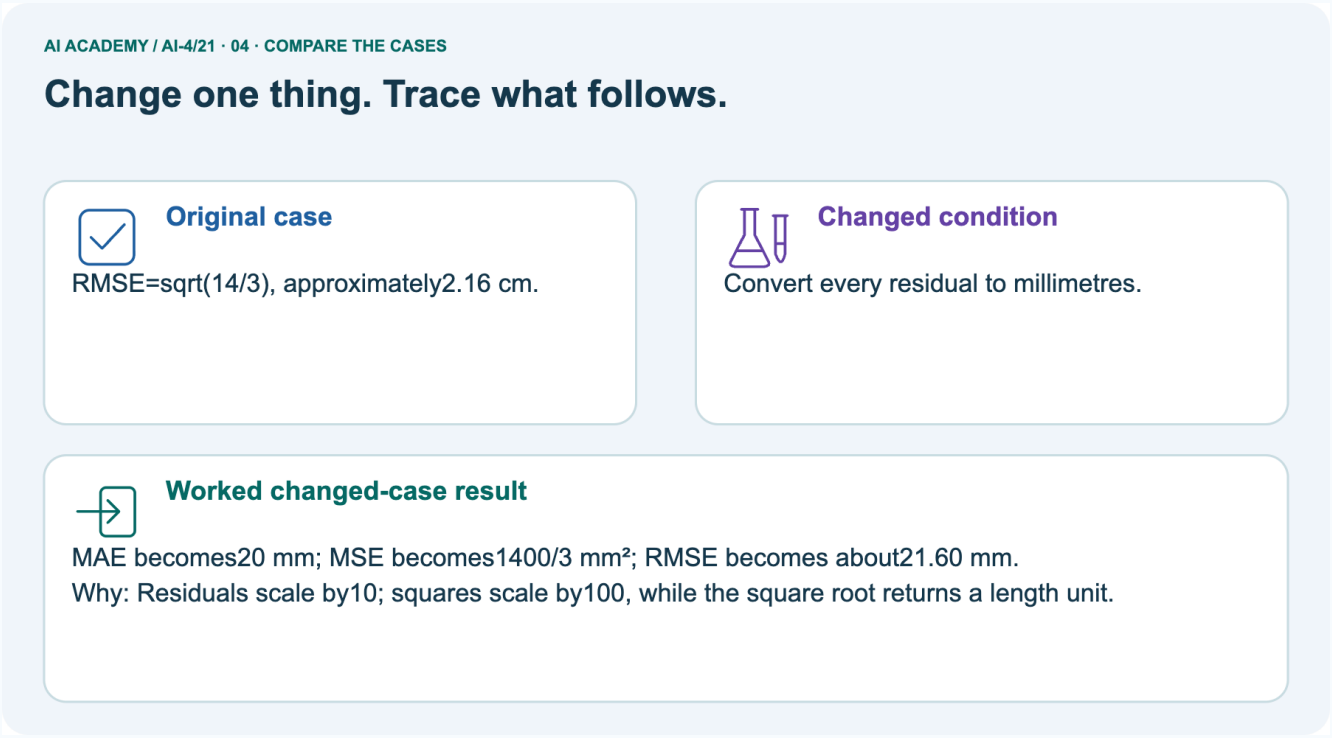
Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

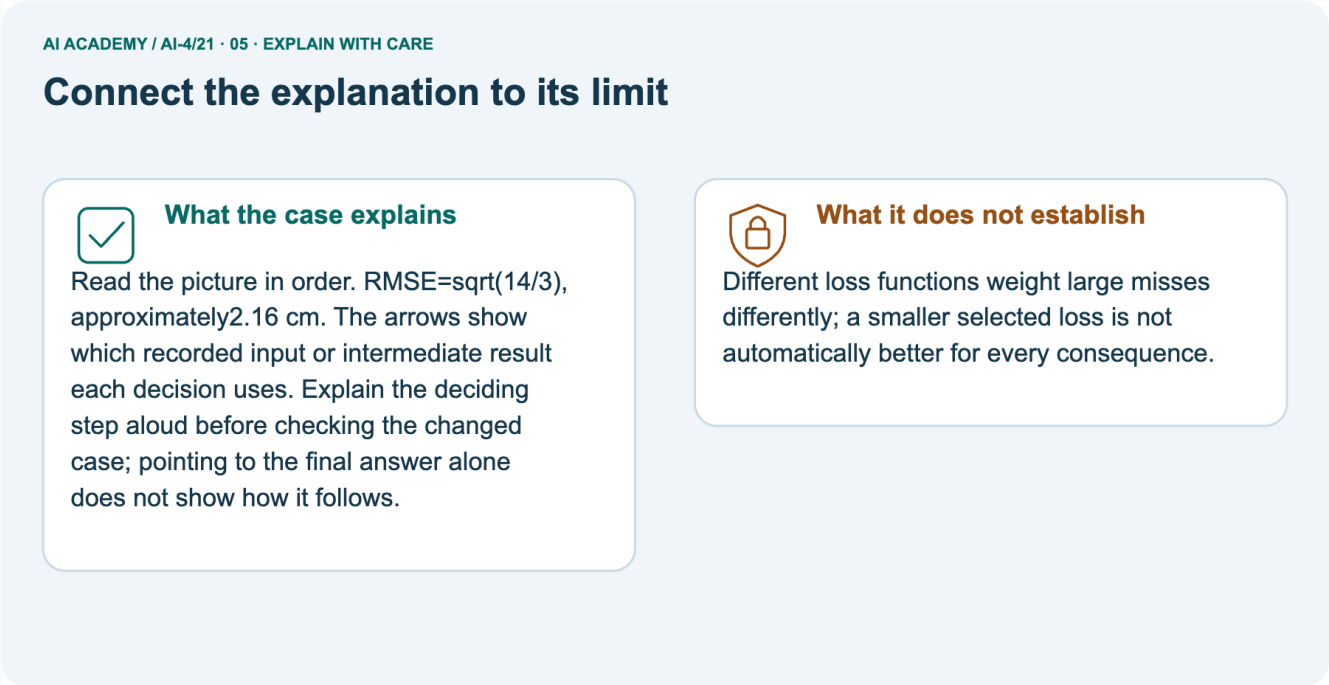
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

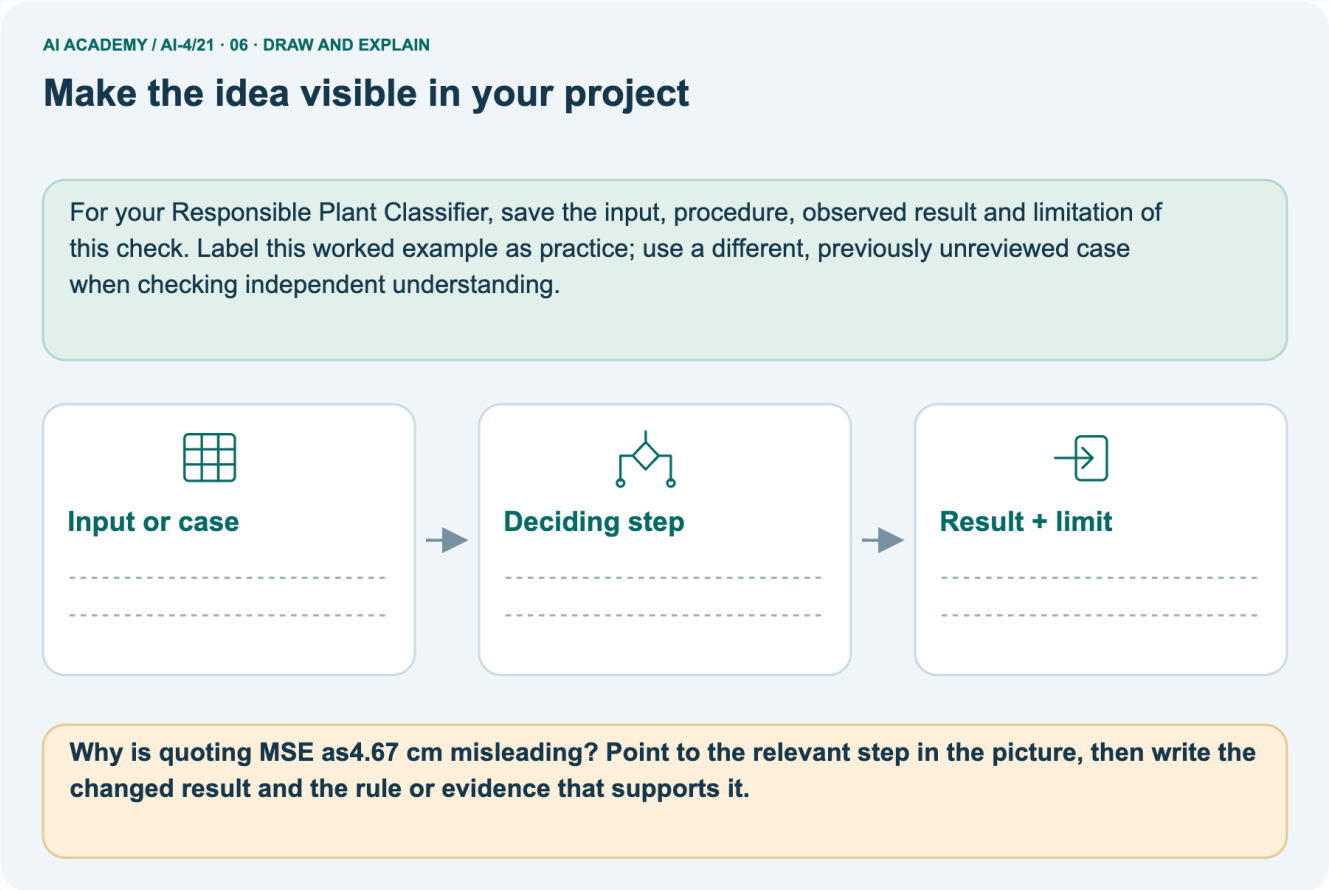
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Its **unit** is cm², not cm. **RMSE** is the corresponding square-root statistic with cm **unit**s. Changed-case check: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.